

QOJWRsR4W2_assign2_deep_learning

Loading Libraries

```
In [1]: # Basic
import os
from os import makedirs
from os import listdir
from shutil import copyfile
from random import seed
from random import random
import numpy as np
import pandas as pd

# visuals
import seaborn as sns
import matplotlib.pyplot as plt
from matplotlib.image import imread
from PIL import Image

# Scikit-learn
from sklearn.model_selection import train_test_split
from sklearn.metrics import classification_report, confusion_matrix, ConfusionMatrixDisplay

# Tensorflow
import tensorflow as tf
from tensorflow.keras.models import Sequential
from tensorflow.keras.preprocessing.image import ImageDataGenerator
from tensorflow.keras.layers import Dense, MaxPooling2D, Dropout, Flatten, BatchNormalization
from tensorflow.keras.callbacks import ReduceLROnPlateau, EarlyStopping
```

```

2025-02-09 15:35:37.838228: I tensorflow/core/util/port.cc:153] oneDNN custom
operations are on. You may see slightly different numerical results due to
floating-point round-off errors from different computation orders. To turn t
hem off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
2025-02-09 15:35:37.839062: I external/local_xla/xla/tsl/cuda/cudart_stub.c
c:32] Could not find cuda drivers on your machine, GPU will not be used.
2025-02-09 15:35:37.841256: I external/local_xla/xla/tsl/cuda/cudart_stub.c
c:32] Could not find cuda drivers on your machine, GPU will not be used.
2025-02-09 15:35:37.848707: E external/local_xla/xla/stream_executor/cuda/cu
da_fft.cc:485] Unable to register cuFFT factory: Attempting to register fact
ory for plugin cuFFT when one has already been registered
2025-02-09 15:35:37.862284: E external/local_xla/xla/stream_executor/cuda/cu
da_dnn.cc:8454] Unable to register cuDNN factory: Attempting to register fac
tory for plugin cuDNN when one has already been registered
2025-02-09 15:35:37.865796: E external/local_xla/xla/stream_executor/cuda/cu
da_blas.cc:1452] Unable to register cuBLAS factory: Attempting to register f
actory for plugin cuBLAS when one has already been registered
2025-02-09 15:35:37.874414: I tensorflow/core/platform/cpu_feature_guard.c
c:210] This TensorFlow binary is optimized to use available CPU instructions
in performance-critical operations.
To enable the following instructions: AVX2 AVX512F AVX512_VNNI FMA, in other
operations, rebuild TensorFlow with the appropriate compiler flags.
2025-02-09 15:35:38.549423: W tensorflow/compiler/tf2tensorrt/utils/py_util
s.cc:38] TF-TRT Warning: Could not find TensorRT

```

Data Extraction

```
In [2]: train_path = "train"
        test_path = "test"
```

```
In [3]: image_dir = "train"

        filenames = os.listdir(image_dir)
        labels = [x.split(".")[0] for x in filenames]

        data = pd.DataFrame({"filename": filenames, "label": labels})

        data.head()
```

```
Out[3]:
```

	filename	label
0	cat.10951.jpg	cat
1	dog.9032.jpg	dog
2	dog.4788.jpg	dog
3	cat.2926.jpg	cat
4	dog.4711.jpg	dog

Data Exploration

We visualize few images of Dogs.

```
In [11]: #First, we start by visualizing the variable of interest.

#Let's view more images in a grid format

plt.figure(figsize=(20,20)) # specifying the overall grid size
plt.subplots_adjust(hspace=0.4)

for i in range(5):

    plt.subplot(1,5,i+1)    # the number of images in the grid is 10*10 (100)
    filename = 'train/' + 'dog.' + str(i) + '.jpg' # take the path
    image = imread(filename)
    plt.imshow(image)
    plt.title('Dog',fontsize=12)
    plt.axis('off')

plt.show()
```



We visualize few images of Cats.

```
In [14]: plt.figure(figsize=(20,20)) # specifying the overall grid size
plt.subplots_adjust(hspace=0.4)

for i in range(5):

    plt.subplot(1,5,i+1)    # the number of images in the grid is 10*10 (100)
    filename = 'train/' + 'cat.' + str(i) + '.jpg'
    image = imread(filename)
    plt.imshow(image)
    plt.title('Cat',fontsize=12)
    plt.axis('off')

plt.show()
```



The images come in different dimensions.

Each time the cell is executed, a new selection of images appears, allowing for an exploration of both categories.

The presence of humans in certain images might pose difficulties for the model in making accurate classifications. In some cases, multiple cats or dogs are present within a single image.

Train Test Split

Using Dataframe

```
In [20]: # train test split using dataframe

labels = data['label']

X_train, X_temp = train_test_split(data, test_size=0.2, stratify=labels, ran

label_test_val = X_temp['label']

X_test, X_val = train_test_split(X_temp, test_size=0.5, stratify=label_test_

print('The shape of train data',X_train.shape)
print('The shape of test data',X_test.shape)
print('The shape of validation data',X_val.shape)
```

```
The shape of train data (20000, 2)
The shape of test data (2500, 2)
The shape of validation data (2500, 2)
```

We will generate a bar plot to visualize the distribution of classes in the training dataset.

This will help us understand how balanced or imbalanced the dataset is, which is important for training a model effectively.

If the dataset has significantly more examples of one class than the other, it could impact the model's ability to generalize well.

```

In [134... labels = ['Cat', 'Dog']

label1, count1 = np.unique(X_train.label, return_counts=True)
label2, count2 = np.unique(X_val.label, return_counts=True)
label3, count3 = np.unique(X_test.label, return_counts=True)

uni1 = pd.DataFrame(data=count1, index=labels, columns=['Count1'])
uni2 = pd.DataFrame(data=count2, index=labels, columns=['Count2'])
uni3 = pd.DataFrame(data=count3, index=labels, columns=['Count3'])

plt.figure(figsize=(20,6), dpi=200)
sns.set_style('darkgrid')

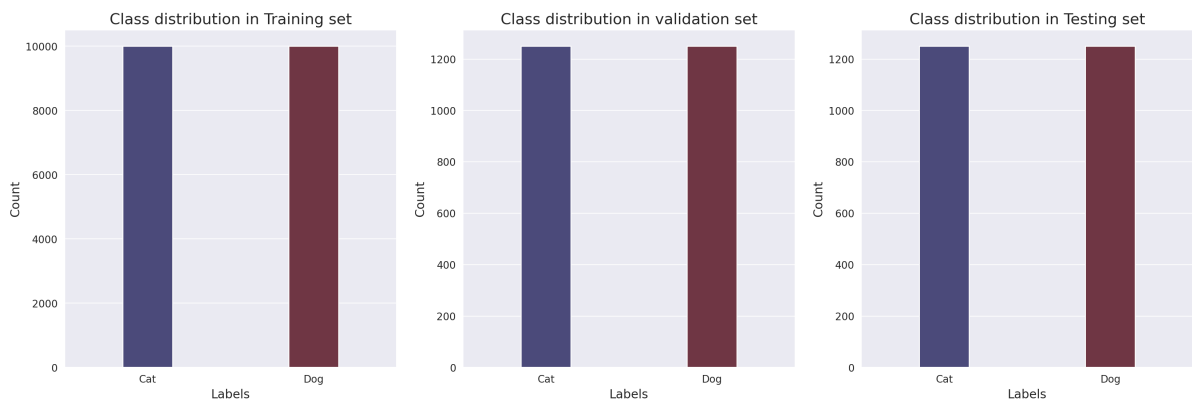
plt.subplot(131)
sns.barplot(data=uni1, x=uni1.index, y='Count1', palette='icefire', width=0.3).s
plt.xlabel('Labels', fontsize=12)
plt.ylabel('Count', fontsize=12)

plt.subplot(132)
sns.barplot(data=uni2, x=uni2.index, y='Count2', palette='icefire', width=0.3).s
plt.xlabel('Labels', fontsize=12)
plt.ylabel('Count', fontsize=12)

plt.subplot(133)
sns.barplot(data=uni3, x=uni3.index, y='Count3', palette='icefire', width=0.3).s
plt.xlabel('Labels', fontsize=12)
plt.ylabel('Count', fontsize=12)

plt.show()

```



Using Directory

```
In [33]: # create directories
dataset_home = 'dataset_dogs_vs_cats/'
subdirs = ['train/', 'test/']

for subdir in subdirs:
    # create label subdirectories
    labldirs = ['dogs/', 'cats/']
    for labldir in labldirs:
        newdir = dataset_home + subdir + labldir
        makedirs(newdir, exist_ok=True)
# seed random number generator
seed(1)
# define ratio of pictures to use for validation
val_ratio = 0.2
# copy training dataset images into subdirectories
src_directory = 'train/'
for file in listdir(src_directory):
    src = src_directory + '/' + file
    dst_dir = 'train/'
    if random() < val_ratio:
        dst_dir = 'test/'
    if file.startswith('cat'):
        dst = dataset_home + dst_dir + 'cats/' + file
        copyfile(src, dst)
    elif file.startswith('dog'):
        dst = dataset_home + dst_dir + 'dogs/' + file
        copyfile(src, dst)

path1 = "/home/student/Documents/Review block/Review block 4/Week 2/Assignme
path2 = "/home/student/Documents/Review block/Review block 4/Week 2/Assignme
path3 = "/home/student/Documents/Review block/Review block 4/Week 2/Assignme
path4 = "/home/student/Documents/Review block/Review block 4/Week 2/Assignme

print('Then number of cat images in training data is' ,len(os.listdir(path1)
print('Then number of dog images in training data is' ,len(os.listdir(path2)
print('Then number of cat images in validation data is' ,len(os.listdir(path
print('Then number of dog images in validation data is' ,len(os.listdir(path
```

```
Then number of cat images in training data is 9924
Then number of dog images in training data is 9986
Then number of cat images in validation data is 2576
Then number of dog images in validation data is 2514
```

Data Preparation

First, we will identify all key parameters and their corresponding values.

These parameters could include class labels, image sizes, data split ratios, and any other relevant features required for analysis.

```
In [34]: # parameters
image_size = 128
image_channel = 3
bat_size = 32
```

Image Data Generator

The data will be loaded using `flow_from_dataframe` and `flow_from_directory` methods.

The batch size is set to 32, and the images will be resized to a consistent size of (128, 128) pixels for processing.

```
In [35]: # Creating image data generator
train_datagen = ImageDataGenerator(rescale=1./255,
                                   rotation_range = 15,
                                   horizontal_flip = True,
                                   zoom_range = 0.2,
                                   shear_range = 0.1,
                                   fill_mode = 'reflect',
                                   width_shift_range = 0.1,
                                   height_shift_range = 0.1)

test_datagen = ImageDataGenerator(rescale=1./255)
```

Using Dataframe

```
In [36]: # Applying image data generator to train and test data

train_generator = train_datagen.flow_from_dataframe(X_train,
                                                    directory = 'train/',
                                                    x_col= 'filename',
                                                    y_col= 'label',
                                                    batch_size = bat_size,
                                                    target_size = (image_size,
                                                                    image_size)
                                                    )

val_generator = test_datagen.flow_from_dataframe(X_val,
                                                  directory = 'train/',
                                                  x_col= 'filename',
                                                  y_col= 'label',
                                                  batch_size = bat_size,
                                                  target_size = (image_size,
                                                                    image_size),
                                                  shuffle=False
                                                  )

test_generator = test_datagen.flow_from_dataframe(X_test,
                                                  directory = 'train/',
                                                  x_col= 'filename',
                                                  y_col= 'label',
                                                  batch_size = bat_size,
                                                  target_size = (image_size,
                                                                    image_size),
                                                  shuffle=False
                                                  )
```

Found 20000 validated image filenames belonging to 2 classes.

Found 2500 validated image filenames belonging to 2 classes.

Found 2500 validated image filenames belonging to 2 classes.

Using Directory

```
In [38]: train_gen = train_datagen.flow_from_directory('/home/student/Documents/Review b
                                                    class_mode='binary',
                                                    target_size = (image_size,imag
                                                    batch_size = bat_size,
                                                    )

val_gen = test_datagen.flow_from_directory('/home/student/Documents/Review b
                                                    class_mode='binary',
                                                    batch_size = bat_size,
                                                    target_size = (image_size,image_si
                                                    shuffle = False
                                                    )
```

```
Found 19910 images belonging to 2 classes.
Found 5090 images belonging to 2 classes.
```

Deep Learning Model

Model Layers

Architecture of the Cat vs Dog Image Classification Model Layers:

Input Layer: A Conv2D layer with 32 filters and ReLU activation.

Convolution Blocks: Three convolutional blocks with increasing filters and ReLU activation.

Batch Normalization & Pooling: Each block includes Batch Normalization, Max Pooling (pool_size=2), and Dropout (rate=0.2).

Fully Connected Layers: A Flatten layer followed by a Dense layer with 512 units and a Dropout layer.

Output Layer: A Dense layer with 2 units and softmax activation.

Components:

Input Layer: Receives images for classification.

Convolutional Layers: Extract key features from images.

Pooling Layers: Reduce spatial dimensions for computational efficiency.

Flatten Layer: Converts feature maps into a 1D vector.

Fully Connected Layers: Perform classification with dense connections.

Output Layer: Produces prediction probabilities for cat and dog classes.

```
In [39]: from tensorflow.keras.applications import ResNet50
from tensorflow.keras.layers import GlobalAveragePooling2D
from tensorflow.keras.models import Model
from tensorflow.keras.layers import Dense, Dropout

def create_resnet_model(input_shape):
    base_model_resnet = ResNet50(weights='imagenet', include_top=False, input_shape=input_shape)
    x = base_model_resnet.output
    x = GlobalAveragePooling2D()(x)
    x = Dense(512, activation='relu')(x)
    x = Dropout(0.5)(x)
    predictions = Dense(2, activation='softmax')(x)
    model_resnet = Model(inputs=base_model_resnet.input, outputs=predictions)
    return model_resnet

model = create_resnet_model((image_size, image_size, image_channel))
model.summary()
```

Downloading data from https://storage.googleapis.com/tensorflow/keras-applications/resnet/resnet50_weights_tf_dim_ordering_tf_kernels_notop.h5

94765736/94765736 ————— 194s 2us/step

Model: "functional"

Layer (type)	Output Shape	Param #	Connected to
input_layer (InputLayer)	(None, 128, 128, 3)	0	-
conv1_pad (ZeroPadding2D)	(None, 134, 134, 3)	0	input_layer[0]
conv1_conv (Conv2D)	(None, 64, 64, 64)	9,472	conv1_pad[0][0]
conv1_bn (BatchNormalizatio...	(None, 64, 64, 64)	256	conv1_conv[0][
conv1_relu (Activation)	(None, 64, 64, 64)	0	conv1_bn[0][0]
pool1_pad (ZeroPadding2D)	(None, 66, 66, 64)	0	conv1_relu[0][
pool1_pool (MaxPooling2D)	(None, 32, 32, 64)	0	pool1_pad[0][0]
conv2_block1_1_conv (Conv2D)	(None, 32, 32, 64)	4,160	pool1_pool[0][
conv2_block1_1_bn (BatchNormalizatio...	(None, 32, 32, 64)	256	conv2_block1_1
conv2_block1_1_relu (Activation)	(None, 32, 32, 64)	0	conv2_block1_1
conv2_block1_2_conv (Conv2D)	(None, 32, 32, 64)	36,928	conv2_block1_1
conv2_block1_2_bn (BatchNormalizatio...	(None, 32, 32, 64)	256	conv2_block1_2
conv2_block1_2_relu (Activation)	(None, 32, 32, 64)	0	conv2_block1_2
conv2_block1_0_conv (Conv2D)	(None, 32, 32, 256)	16,640	pool1_pool[0][
conv2_block1_3_conv (Conv2D)	(None, 32, 32, 256)	16,640	conv2_block1_2
conv2_block1_0_bn (BatchNormalizatio...	(None, 32, 32, 256)	1,024	conv2_block1_0
conv2_block1_3_bn (BatchNormalizatio...	(None, 32, 32, 256)	1,024	conv2_block1_3
conv2_block1_add (Add)	(None, 32, 32, 256)	0	conv2_block1_0 conv2_block1_3

conv2_block1_out (Activation)	(None, 32, 32, 256)	0	conv2_block1_a
conv2_block2_1_conv (Conv2D)	(None, 32, 32, 64)	16,448	conv2_block1_o
conv2_block2_1_bn (BatchNormalizatio...	(None, 32, 32, 64)	256	conv2_block2_1
conv2_block2_1_relu (Activation)	(None, 32, 32, 64)	0	conv2_block2_1
conv2_block2_2_conv (Conv2D)	(None, 32, 32, 64)	36,928	conv2_block2_1
conv2_block2_2_bn (BatchNormalizatio...	(None, 32, 32, 64)	256	conv2_block2_2
conv2_block2_2_relu (Activation)	(None, 32, 32, 64)	0	conv2_block2_2
conv2_block2_3_conv (Conv2D)	(None, 32, 32, 256)	16,640	conv2_block2_2
conv2_block2_3_bn (BatchNormalizatio...	(None, 32, 32, 256)	1,024	conv2_block2_3
conv2_block2_add (Add)	(None, 32, 32, 256)	0	conv2_block1_o conv2_block2_3
conv2_block2_out (Activation)	(None, 32, 32, 256)	0	conv2_block2_a
conv2_block3_1_conv (Conv2D)	(None, 32, 32, 64)	16,448	conv2_block2_o
conv2_block3_1_bn (BatchNormalizatio...	(None, 32, 32, 64)	256	conv2_block3_1
conv2_block3_1_relu (Activation)	(None, 32, 32, 64)	0	conv2_block3_1
conv2_block3_2_conv (Conv2D)	(None, 32, 32, 64)	36,928	conv2_block3_1
conv2_block3_2_bn (BatchNormalizatio...	(None, 32, 32, 64)	256	conv2_block3_2
conv2_block3_2_relu (Activation)	(None, 32, 32, 64)	0	conv2_block3_2
conv2_block3_3_conv (Conv2D)	(None, 32, 32, 256)	16,640	conv2_block3_2
conv2_block3_3_bn (BatchNormalizatio...	(None, 32, 32, 256)	1,024	conv2_block3_3

conv2_block3_add (Add)	(None, 32, 32, 256)	0	conv2_block2_o conv2_block3_3
conv2_block3_out (Activation)	(None, 32, 32, 256)	0	conv2_block3_a
conv3_block1_1_conv (Conv2D)	(None, 16, 16, 128)	32,896	conv2_block3_o
conv3_block1_1_bn (BatchNormalizatio...	(None, 16, 16, 128)	512	conv3_block1_1
conv3_block1_1_relu (Activation)	(None, 16, 16, 128)	0	conv3_block1_1
conv3_block1_2_conv (Conv2D)	(None, 16, 16, 128)	147,584	conv3_block1_1
conv3_block1_2_bn (BatchNormalizatio...	(None, 16, 16, 128)	512	conv3_block1_2
conv3_block1_2_relu (Activation)	(None, 16, 16, 128)	0	conv3_block1_2
conv3_block1_0_conv (Conv2D)	(None, 16, 16, 512)	131,584	conv2_block3_o
conv3_block1_3_conv (Conv2D)	(None, 16, 16, 512)	66,048	conv3_block1_2
conv3_block1_0_bn (BatchNormalizatio...	(None, 16, 16, 512)	2,048	conv3_block1_0
conv3_block1_3_bn (BatchNormalizatio...	(None, 16, 16, 512)	2,048	conv3_block1_3
conv3_block1_add (Add)	(None, 16, 16, 512)	0	conv3_block1_0 conv3_block1_3
conv3_block1_out (Activation)	(None, 16, 16, 512)	0	conv3_block1_a
conv3_block2_1_conv (Conv2D)	(None, 16, 16, 128)	65,664	conv3_block1_o
conv3_block2_1_bn (BatchNormalizatio...	(None, 16, 16, 128)	512	conv3_block2_1
conv3_block2_1_relu (Activation)	(None, 16, 16, 128)	0	conv3_block2_1
conv3_block2_2_conv (Conv2D)	(None, 16, 16, 128)	147,584	conv3_block2_1
conv3_block2_2_bn (BatchNormalizatio...	(None, 16, 16, 128)	512	conv3_block2_2

conv3_block2_2_relu (Activation)	(None, 16, 16, 128)	0	conv3_block2_2
conv3_block2_3_conv (Conv2D)	(None, 16, 16, 512)	66,048	conv3_block2_2
conv3_block2_3_bn (BatchNormalizatio...	(None, 16, 16, 512)	2,048	conv3_block2_3
conv3_block2_add (Add)	(None, 16, 16, 512)	0	conv3_block1_o conv3_block2_3
conv3_block2_out (Activation)	(None, 16, 16, 512)	0	conv3_block2_a
conv3_block3_1_conv (Conv2D)	(None, 16, 16, 128)	65,664	conv3_block2_o
conv3_block3_1_bn (BatchNormalizatio...	(None, 16, 16, 128)	512	conv3_block3_1
conv3_block3_1_relu (Activation)	(None, 16, 16, 128)	0	conv3_block3_1
conv3_block3_2_conv (Conv2D)	(None, 16, 16, 128)	147,584	conv3_block3_1
conv3_block3_2_bn (BatchNormalizatio...	(None, 16, 16, 128)	512	conv3_block3_2
conv3_block3_2_relu (Activation)	(None, 16, 16, 128)	0	conv3_block3_2
conv3_block3_3_conv (Conv2D)	(None, 16, 16, 512)	66,048	conv3_block3_2
conv3_block3_3_bn (BatchNormalizatio...	(None, 16, 16, 512)	2,048	conv3_block3_3
conv3_block3_add (Add)	(None, 16, 16, 512)	0	conv3_block2_o conv3_block3_3
conv3_block3_out (Activation)	(None, 16, 16, 512)	0	conv3_block3_a
conv3_block4_1_conv (Conv2D)	(None, 16, 16, 128)	65,664	conv3_block3_o
conv3_block4_1_bn (BatchNormalizatio...	(None, 16, 16, 128)	512	conv3_block4_1
conv3_block4_1_relu (Activation)	(None, 16, 16, 128)	0	conv3_block4_1
conv3_block4_2_conv (Conv2D)	(None, 16, 16, 128)	147,584	conv3_block4_1

conv3_block4_2_bn (BatchNormalizatio...	(None, 16, 16, 128)	512	conv3_block4_2
conv3_block4_2_relu (Activation)	(None, 16, 16, 128)	0	conv3_block4_2
conv3_block4_3_conv (Conv2D)	(None, 16, 16, 512)	66,048	conv3_block4_2
conv3_block4_3_bn (BatchNormalizatio...	(None, 16, 16, 512)	2,048	conv3_block4_3
conv3_block4_add (Add)	(None, 16, 16, 512)	0	conv3_block3_o conv3_block4_3
conv3_block4_out (Activation)	(None, 16, 16, 512)	0	conv3_block4_a
conv4_block1_1_conv (Conv2D)	(None, 8, 8, 256)	131,328	conv3_block4_o
conv4_block1_1_bn (BatchNormalizatio...	(None, 8, 8, 256)	1,024	conv4_block1_1
conv4_block1_1_relu (Activation)	(None, 8, 8, 256)	0	conv4_block1_1
conv4_block1_2_conv (Conv2D)	(None, 8, 8, 256)	590,080	conv4_block1_1
conv4_block1_2_bn (BatchNormalizatio...	(None, 8, 8, 256)	1,024	conv4_block1_2
conv4_block1_2_relu (Activation)	(None, 8, 8, 256)	0	conv4_block1_2
conv4_block1_0_conv (Conv2D)	(None, 8, 8, 1024)	525,312	conv3_block4_o
conv4_block1_3_conv (Conv2D)	(None, 8, 8, 1024)	263,168	conv4_block1_2
conv4_block1_0_bn (BatchNormalizatio...	(None, 8, 8, 1024)	4,096	conv4_block1_0
conv4_block1_3_bn (BatchNormalizatio...	(None, 8, 8, 1024)	4,096	conv4_block1_3
conv4_block1_add (Add)	(None, 8, 8, 1024)	0	conv4_block1_0 conv4_block1_3
conv4_block1_out (Activation)	(None, 8, 8, 1024)	0	conv4_block1_a
conv4_block2_1_conv (Conv2D)	(None, 8, 8, 256)	262,400	conv4_block1_o

conv4_block2_1_bn (BatchNormalizatio...	(None, 8, 8, 256)	1,024	conv4_block2_1
conv4_block2_1_relu (Activation)	(None, 8, 8, 256)	0	conv4_block2_1
conv4_block2_2_conv (Conv2D)	(None, 8, 8, 256)	590,080	conv4_block2_1
conv4_block2_2_bn (BatchNormalizatio...	(None, 8, 8, 256)	1,024	conv4_block2_2
conv4_block2_2_relu (Activation)	(None, 8, 8, 256)	0	conv4_block2_2
conv4_block2_3_conv (Conv2D)	(None, 8, 8, 1024)	263,168	conv4_block2_2
conv4_block2_3_bn (BatchNormalizatio...	(None, 8, 8, 1024)	4,096	conv4_block2_3
conv4_block2_add (Add)	(None, 8, 8, 1024)	0	conv4_block1_o conv4_block2_3
conv4_block2_out (Activation)	(None, 8, 8, 1024)	0	conv4_block2_a
conv4_block3_1_conv (Conv2D)	(None, 8, 8, 256)	262,400	conv4_block2_o
conv4_block3_1_bn (BatchNormalizatio...	(None, 8, 8, 256)	1,024	conv4_block3_1
conv4_block3_1_relu (Activation)	(None, 8, 8, 256)	0	conv4_block3_1
conv4_block3_2_conv (Conv2D)	(None, 8, 8, 256)	590,080	conv4_block3_1
conv4_block3_2_bn (BatchNormalizatio...	(None, 8, 8, 256)	1,024	conv4_block3_2
conv4_block3_2_relu (Activation)	(None, 8, 8, 256)	0	conv4_block3_2
conv4_block3_3_conv (Conv2D)	(None, 8, 8, 1024)	263,168	conv4_block3_2
conv4_block3_3_bn (BatchNormalizatio...	(None, 8, 8, 1024)	4,096	conv4_block3_3
conv4_block3_add (Add)	(None, 8, 8, 1024)	0	conv4_block2_o conv4_block3_3
conv4_block3_out (Activation)	(None, 8, 8, 1024)	0	conv4_block3_a

conv4_block4_1_conv (Conv2D)	(None, 8, 8, 256)	262,400	conv4_block3_o
conv4_block4_1_bn (BatchNormalizatio...	(None, 8, 8, 256)	1,024	conv4_block4_1
conv4_block4_1_relu (Activation)	(None, 8, 8, 256)	0	conv4_block4_1
conv4_block4_2_conv (Conv2D)	(None, 8, 8, 256)	590,080	conv4_block4_1
conv4_block4_2_bn (BatchNormalizatio...	(None, 8, 8, 256)	1,024	conv4_block4_2
conv4_block4_2_relu (Activation)	(None, 8, 8, 256)	0	conv4_block4_2
conv4_block4_3_conv (Conv2D)	(None, 8, 8, 1024)	263,168	conv4_block4_2
conv4_block4_3_bn (BatchNormalizatio...	(None, 8, 8, 1024)	4,096	conv4_block4_3
conv4_block4_add (Add)	(None, 8, 8, 1024)	0	conv4_block3_o conv4_block4_3
conv4_block4_out (Activation)	(None, 8, 8, 1024)	0	conv4_block4_a
conv4_block5_1_conv (Conv2D)	(None, 8, 8, 256)	262,400	conv4_block4_o
conv4_block5_1_bn (BatchNormalizatio...	(None, 8, 8, 256)	1,024	conv4_block5_1
conv4_block5_1_relu (Activation)	(None, 8, 8, 256)	0	conv4_block5_1
conv4_block5_2_conv (Conv2D)	(None, 8, 8, 256)	590,080	conv4_block5_1
conv4_block5_2_bn (BatchNormalizatio...	(None, 8, 8, 256)	1,024	conv4_block5_2
conv4_block5_2_relu (Activation)	(None, 8, 8, 256)	0	conv4_block5_2
conv4_block5_3_conv (Conv2D)	(None, 8, 8, 1024)	263,168	conv4_block5_2
conv4_block5_3_bn (BatchNormalizatio...	(None, 8, 8, 1024)	4,096	conv4_block5_3
conv4_block5_add (Add)	(None, 8, 8, 1024)	0	conv4_block4_o conv4_block5_3

conv4_block5_out (Activation)	(None, 8, 8, 1024)	0	conv4_block5_a
conv4_block6_1_conv (Conv2D)	(None, 8, 8, 256)	262,400	conv4_block5_o
conv4_block6_1_bn (BatchNormalizatio...	(None, 8, 8, 256)	1,024	conv4_block6_1
conv4_block6_1_relu (Activation)	(None, 8, 8, 256)	0	conv4_block6_1
conv4_block6_2_conv (Conv2D)	(None, 8, 8, 256)	590,080	conv4_block6_1
conv4_block6_2_bn (BatchNormalizatio...	(None, 8, 8, 256)	1,024	conv4_block6_2
conv4_block6_2_relu (Activation)	(None, 8, 8, 256)	0	conv4_block6_2
conv4_block6_3_conv (Conv2D)	(None, 8, 8, 1024)	263,168	conv4_block6_2
conv4_block6_3_bn (BatchNormalizatio...	(None, 8, 8, 1024)	4,096	conv4_block6_3
conv4_block6_add (Add)	(None, 8, 8, 1024)	0	conv4_block5_o conv4_block6_3
conv4_block6_out (Activation)	(None, 8, 8, 1024)	0	conv4_block6_a
conv5_block1_1_conv (Conv2D)	(None, 4, 4, 512)	524,800	conv4_block6_o
conv5_block1_1_bn (BatchNormalizatio...	(None, 4, 4, 512)	2,048	conv5_block1_1
conv5_block1_1_relu (Activation)	(None, 4, 4, 512)	0	conv5_block1_1
conv5_block1_2_conv (Conv2D)	(None, 4, 4, 512)	2,359,808	conv5_block1_1
conv5_block1_2_bn (BatchNormalizatio...	(None, 4, 4, 512)	2,048	conv5_block1_2
conv5_block1_2_relu (Activation)	(None, 4, 4, 512)	0	conv5_block1_2
conv5_block1_0_conv (Conv2D)	(None, 4, 4, 2048)	2,099,200	conv4_block6_o
conv5_block1_3_conv (Conv2D)	(None, 4, 4, 2048)	1,050,624	conv5_block1_2

conv5_block1_0_bn (BatchNormalizatio...	(None, 4, 4, 2048)	8,192	conv5_block1_0
conv5_block1_3_bn (BatchNormalizatio...	(None, 4, 4, 2048)	8,192	conv5_block1_3
conv5_block1_add (Add)	(None, 4, 4, 2048)	0	conv5_block1_0 conv5_block1_3
conv5_block1_out (Activation)	(None, 4, 4, 2048)	0	conv5_block1_a
conv5_block2_1_conv (Conv2D)	(None, 4, 4, 512)	1,049,088	conv5_block1_o
conv5_block2_1_bn (BatchNormalizatio...	(None, 4, 4, 512)	2,048	conv5_block2_1
conv5_block2_1_relu (Activation)	(None, 4, 4, 512)	0	conv5_block2_1
conv5_block2_2_conv (Conv2D)	(None, 4, 4, 512)	2,359,808	conv5_block2_1
conv5_block2_2_bn (BatchNormalizatio...	(None, 4, 4, 512)	2,048	conv5_block2_2
conv5_block2_2_relu (Activation)	(None, 4, 4, 512)	0	conv5_block2_2
conv5_block2_3_conv (Conv2D)	(None, 4, 4, 2048)	1,050,624	conv5_block2_2
conv5_block2_3_bn (BatchNormalizatio...	(None, 4, 4, 2048)	8,192	conv5_block2_3
conv5_block2_add (Add)	(None, 4, 4, 2048)	0	conv5_block1_o conv5_block2_3
conv5_block2_out (Activation)	(None, 4, 4, 2048)	0	conv5_block2_a
conv5_block3_1_conv (Conv2D)	(None, 4, 4, 512)	1,049,088	conv5_block2_o
conv5_block3_1_bn (BatchNormalizatio...	(None, 4, 4, 512)	2,048	conv5_block3_1
conv5_block3_1_relu (Activation)	(None, 4, 4, 512)	0	conv5_block3_1
conv5_block3_2_conv (Conv2D)	(None, 4, 4, 512)	2,359,808	conv5_block3_1
conv5_block3_2_bn (BatchNormalizatio...	(None, 4, 4, 512)	2,048	conv5_block3_2

conv5_block3_2_relu (Activation)	(None, 4, 4, 512)	0	conv5_block3_2
conv5_block3_3_conv (Conv2D)	(None, 4, 4, 2048)	1,050,624	conv5_block3_2
conv5_block3_3_bn (BatchNormalizatio...	(None, 4, 4, 2048)	8,192	conv5_block3_3
conv5_block3_add (Add)	(None, 4, 4, 2048)	0	conv5_block2_o conv5_block3_3
conv5_block3_out (Activation)	(None, 4, 4, 2048)	0	conv5_block3_a
global_average_poo... (GlobalAveragePool...	(None, 2048)	0	conv5_block3_o
dense (Dense)	(None, 512)	1,049,088	global_average
dropout (Dropout)	(None, 512)	0	dense[0][0]
dense_1 (Dense)	(None, 2)	1,026	dropout[0][0]

Total params: 24,637,826 (93.99 MB)

Trainable params: 24,584,706 (93.78 MB)

Non-trainable params: 53,120 (207.50 KB)

```
In [40]: from tensorflow.keras.applications import DenseNet121
from tensorflow.keras.layers import GlobalAveragePooling2D
from tensorflow.keras.models import Model
from tensorflow.keras.layers import Dense, Dropout

def create_densenet_model(input_shape):
    base_model_densenet = DenseNet121(weights='imagenet', include_top=False,
    x = base_model_densenet.output
    x = GlobalAveragePooling2D()(x)
    x = Dense(512, activation='relu')(x)
    x = Dropout(0.5)(x)
    predictions = Dense(2, activation='softmax')(x)
    model_densenet = Model(inputs=base_model_densenet.input, outputs=predictions)
    return model_densenet

model = create_densenet_model((image_size, image_size, image_channel))
model.summary()
```

Downloading data from https://storage.googleapis.com/tensorflow/keras-applications/densenet/densenet121_weights_tf_dim_ordering_tf_kernels_notop.h5

29084464/29084464 ————— 46s 2us/step

Model: "functional_1"

Layer (type)	Output Shape	Param #	Connected to
input_layer_1 (InputLayer)	(None, 128, 128, 3)	0	-
zero_padding2d (ZeroPadding2D)	(None, 134, 134, 3)	0	input_layer_1[
conv1_conv (Conv2D)	(None, 64, 64, 64)	9,408	zero_padding2d
conv1_bn (BatchNormalizatio...	(None, 64, 64, 64)	256	conv1_conv[0][
conv1_relu (Activation)	(None, 64, 64, 64)	0	conv1_bn[0][0]
zero_padding2d_1 (ZeroPadding2D)	(None, 66, 66, 64)	0	conv1_relu[0][
pool1 (MaxPooling2D)	(None, 32, 32, 64)	0	zero_padding2d
conv2_block1_0_bn (BatchNormalizatio...	(None, 32, 32, 64)	256	pool1[0][0]
conv2_block1_0_relu (Activation)	(None, 32, 32, 64)	0	conv2_block1_0
conv2_block1_1_conv (Conv2D)	(None, 32, 32, 128)	8,192	conv2_block1_0
conv2_block1_1_bn (BatchNormalizatio...	(None, 32, 32, 128)	512	conv2_block1_1
conv2_block1_1_relu (Activation)	(None, 32, 32, 128)	0	conv2_block1_1
conv2_block1_2_conv (Conv2D)	(None, 32, 32, 32)	36,864	conv2_block1_1
conv2_block1_concat (Concatenate)	(None, 32, 32, 96)	0	pool1[0][0], conv2_block1_2
conv2_block2_0_bn (BatchNormalizatio...	(None, 32, 32, 96)	384	conv2_block1_c
conv2_block2_0_relu (Activation)	(None, 32, 32, 96)	0	conv2_block2_0
conv2_block2_1_conv (Conv2D)	(None, 32, 32, 128)	12,288	conv2_block2_0
conv2_block2_1_bn (BatchNormalizatio...	(None, 32, 32, 128)	512	conv2_block2_1

conv2_block2_1_relu (Activation)	(None, 32, 32, 128)	0	conv2_block2_1
conv2_block2_2_conv (Conv2D)	(None, 32, 32, 32)	36,864	conv2_block2_1
conv2_block2_concat (Concatenate)	(None, 32, 32, 128)	0	conv2_block1_c conv2_block2_2
conv2_block3_0_bn (BatchNormalizatio...	(None, 32, 32, 128)	512	conv2_block2_c
conv2_block3_0_relu (Activation)	(None, 32, 32, 128)	0	conv2_block3_0
conv2_block3_1_conv (Conv2D)	(None, 32, 32, 128)	16,384	conv2_block3_0
conv2_block3_1_bn (BatchNormalizatio...	(None, 32, 32, 128)	512	conv2_block3_1
conv2_block3_1_relu (Activation)	(None, 32, 32, 128)	0	conv2_block3_1
conv2_block3_2_conv (Conv2D)	(None, 32, 32, 32)	36,864	conv2_block3_1
conv2_block3_concat (Concatenate)	(None, 32, 32, 160)	0	conv2_block2_c conv2_block3_2
conv2_block4_0_bn (BatchNormalizatio...	(None, 32, 32, 160)	640	conv2_block3_c
conv2_block4_0_relu (Activation)	(None, 32, 32, 160)	0	conv2_block4_0
conv2_block4_1_conv (Conv2D)	(None, 32, 32, 128)	20,480	conv2_block4_0
conv2_block4_1_bn (BatchNormalizatio...	(None, 32, 32, 128)	512	conv2_block4_1
conv2_block4_1_relu (Activation)	(None, 32, 32, 128)	0	conv2_block4_1
conv2_block4_2_conv (Conv2D)	(None, 32, 32, 32)	36,864	conv2_block4_1
conv2_block4_concat (Concatenate)	(None, 32, 32, 192)	0	conv2_block3_c conv2_block4_2
conv2_block5_0_bn (BatchNormalizatio...	(None, 32, 32, 192)	768	conv2_block4_c
conv2_block5_0_relu (Activation)	(None, 32, 32, 192)	0	conv2_block5_0

conv2_block5_1_conv (Conv2D)	(None, 32, 32, 128)	24,576	conv2_block5_0
conv2_block5_1_bn (BatchNormalizatio...	(None, 32, 32, 128)	512	conv2_block5_1
conv2_block5_1_relu (Activation)	(None, 32, 32, 128)	0	conv2_block5_1
conv2_block5_2_conv (Conv2D)	(None, 32, 32, 32)	36,864	conv2_block5_1
conv2_block5_concat (Concatenate)	(None, 32, 32, 224)	0	conv2_block4_c conv2_block5_2
conv2_block6_0_bn (BatchNormalizatio...	(None, 32, 32, 224)	896	conv2_block5_c
conv2_block6_0_relu (Activation)	(None, 32, 32, 224)	0	conv2_block6_0
conv2_block6_1_conv (Conv2D)	(None, 32, 32, 128)	28,672	conv2_block6_0
conv2_block6_1_bn (BatchNormalizatio...	(None, 32, 32, 128)	512	conv2_block6_1
conv2_block6_1_relu (Activation)	(None, 32, 32, 128)	0	conv2_block6_1
conv2_block6_2_conv (Conv2D)	(None, 32, 32, 32)	36,864	conv2_block6_1
conv2_block6_concat (Concatenate)	(None, 32, 32, 256)	0	conv2_block5_c conv2_block6_2
pool2_bn (BatchNormalizatio...	(None, 32, 32, 256)	1,024	conv2_block6_c
pool2_relu (Activation)	(None, 32, 32, 256)	0	pool2_bn[0][0]
pool2_conv (Conv2D)	(None, 32, 32, 128)	32,768	pool2_relu[0][
pool2_pool (AveragePooling2D)	(None, 16, 16, 128)	0	pool2_conv[0][
conv3_block1_0_bn (BatchNormalizatio...	(None, 16, 16, 128)	512	pool2_pool[0][
conv3_block1_0_relu (Activation)	(None, 16, 16, 128)	0	conv3_block1_0
conv3_block1_1_conv (Conv2D)	(None, 16, 16, 128)	16,384	conv3_block1_0

conv3_block1_1_bn (BatchNormalizatio...	(None, 16, 16, 128)	512	conv3_block1_1
conv3_block1_1_relu (Activation)	(None, 16, 16, 128)	0	conv3_block1_1
conv3_block1_2_conv (Conv2D)	(None, 16, 16, 32)	36,864	conv3_block1_1
conv3_block1_concat (Concatenate)	(None, 16, 16, 160)	0	pool2_pool[0][conv3_block1_2
conv3_block2_0_bn (BatchNormalizatio...	(None, 16, 16, 160)	640	conv3_block1_c
conv3_block2_0_relu (Activation)	(None, 16, 16, 160)	0	conv3_block2_0
conv3_block2_1_conv (Conv2D)	(None, 16, 16, 128)	20,480	conv3_block2_0
conv3_block2_1_bn (BatchNormalizatio...	(None, 16, 16, 128)	512	conv3_block2_1
conv3_block2_1_relu (Activation)	(None, 16, 16, 128)	0	conv3_block2_1
conv3_block2_2_conv (Conv2D)	(None, 16, 16, 32)	36,864	conv3_block2_1
conv3_block2_concat (Concatenate)	(None, 16, 16, 192)	0	conv3_block1_c conv3_block2_2
conv3_block3_0_bn (BatchNormalizatio...	(None, 16, 16, 192)	768	conv3_block2_c
conv3_block3_0_relu (Activation)	(None, 16, 16, 192)	0	conv3_block3_0
conv3_block3_1_conv (Conv2D)	(None, 16, 16, 128)	24,576	conv3_block3_0
conv3_block3_1_bn (BatchNormalizatio...	(None, 16, 16, 128)	512	conv3_block3_1
conv3_block3_1_relu (Activation)	(None, 16, 16, 128)	0	conv3_block3_1
conv3_block3_2_conv (Conv2D)	(None, 16, 16, 32)	36,864	conv3_block3_1
conv3_block3_concat (Concatenate)	(None, 16, 16, 224)	0	conv3_block2_c conv3_block3_2
conv3_block4_0_bn (BatchNormalizatio...	(None, 16, 16, 224)	896	conv3_block3_c

conv3_block4_0_relu (Activation)	(None, 16, 16, 224)	0	conv3_block4_0
conv3_block4_1_conv (Conv2D)	(None, 16, 16, 128)	28,672	conv3_block4_0
conv3_block4_1_bn (BatchNormalizatio...	(None, 16, 16, 128)	512	conv3_block4_1
conv3_block4_1_relu (Activation)	(None, 16, 16, 128)	0	conv3_block4_1
conv3_block4_2_conv (Conv2D)	(None, 16, 16, 32)	36,864	conv3_block4_1
conv3_block4_concat (Concatenate)	(None, 16, 16, 256)	0	conv3_block3_c conv3_block4_2
conv3_block5_0_bn (BatchNormalizatio...	(None, 16, 16, 256)	1,024	conv3_block4_c
conv3_block5_0_relu (Activation)	(None, 16, 16, 256)	0	conv3_block5_0
conv3_block5_1_conv (Conv2D)	(None, 16, 16, 128)	32,768	conv3_block5_0
conv3_block5_1_bn (BatchNormalizatio...	(None, 16, 16, 128)	512	conv3_block5_1
conv3_block5_1_relu (Activation)	(None, 16, 16, 128)	0	conv3_block5_1
conv3_block5_2_conv (Conv2D)	(None, 16, 16, 32)	36,864	conv3_block5_1
conv3_block5_concat (Concatenate)	(None, 16, 16, 288)	0	conv3_block4_c conv3_block5_2
conv3_block6_0_bn (BatchNormalizatio...	(None, 16, 16, 288)	1,152	conv3_block5_c
conv3_block6_0_relu (Activation)	(None, 16, 16, 288)	0	conv3_block6_0
conv3_block6_1_conv (Conv2D)	(None, 16, 16, 128)	36,864	conv3_block6_0
conv3_block6_1_bn (BatchNormalizatio...	(None, 16, 16, 128)	512	conv3_block6_1
conv3_block6_1_relu (Activation)	(None, 16, 16, 128)	0	conv3_block6_1
conv3_block6_2_conv (Conv2D)	(None, 16, 16, 32)	36,864	conv3_block6_1

conv3_block6_concat (Concatenate)	(None, 16, 16, 320)	0	conv3_block5_c conv3_block6_2
conv3_block7_0_bn (BatchNormalizatio...	(None, 16, 16, 320)	1,280	conv3_block6_c
conv3_block7_0_relu (Activation)	(None, 16, 16, 320)	0	conv3_block7_0
conv3_block7_1_conv (Conv2D)	(None, 16, 16, 128)	40,960	conv3_block7_0
conv3_block7_1_bn (BatchNormalizatio...	(None, 16, 16, 128)	512	conv3_block7_1
conv3_block7_1_relu (Activation)	(None, 16, 16, 128)	0	conv3_block7_1
conv3_block7_2_conv (Conv2D)	(None, 16, 16, 32)	36,864	conv3_block7_1
conv3_block7_concat (Concatenate)	(None, 16, 16, 352)	0	conv3_block6_c conv3_block7_2
conv3_block8_0_bn (BatchNormalizatio...	(None, 16, 16, 352)	1,408	conv3_block7_c
conv3_block8_0_relu (Activation)	(None, 16, 16, 352)	0	conv3_block8_0
conv3_block8_1_conv (Conv2D)	(None, 16, 16, 128)	45,056	conv3_block8_0
conv3_block8_1_bn (BatchNormalizatio...	(None, 16, 16, 128)	512	conv3_block8_1
conv3_block8_1_relu (Activation)	(None, 16, 16, 128)	0	conv3_block8_1
conv3_block8_2_conv (Conv2D)	(None, 16, 16, 32)	36,864	conv3_block8_1
conv3_block8_concat (Concatenate)	(None, 16, 16, 384)	0	conv3_block7_c conv3_block8_2
conv3_block9_0_bn (BatchNormalizatio...	(None, 16, 16, 384)	1,536	conv3_block8_c
conv3_block9_0_relu (Activation)	(None, 16, 16, 384)	0	conv3_block9_0
conv3_block9_1_conv (Conv2D)	(None, 16, 16, 128)	49,152	conv3_block9_0
conv3_block9_1_bn (BatchNormalizatio...	(None, 16, 16, 128)	512	conv3_block9_1

conv3_block9_1_relu (Activation)	(None, 16, 16, 128)	0	conv3_block9_1
conv3_block9_2_conv (Conv2D)	(None, 16, 16, 32)	36,864	conv3_block9_1
conv3_block9_concat (Concatenate)	(None, 16, 16, 416)	0	conv3_block8_c conv3_block9_2
conv3_block10_0_bn (BatchNormalizatio...	(None, 16, 16, 416)	1,664	conv3_block9_c
conv3_block10_0_re... (Activation)	(None, 16, 16, 416)	0	conv3_block10_
conv3_block10_1_co... (Conv2D)	(None, 16, 16, 128)	53,248	conv3_block10_
conv3_block10_1_bn (BatchNormalizatio...	(None, 16, 16, 128)	512	conv3_block10_
conv3_block10_1_re... (Activation)	(None, 16, 16, 128)	0	conv3_block10_
conv3_block10_2_co... (Conv2D)	(None, 16, 16, 32)	36,864	conv3_block10_
conv3_block10_conc... (Concatenate)	(None, 16, 16, 448)	0	conv3_block9_c conv3_block10_
conv3_block11_0_bn (BatchNormalizatio...	(None, 16, 16, 448)	1,792	conv3_block10_
conv3_block11_0_re... (Activation)	(None, 16, 16, 448)	0	conv3_block11_
conv3_block11_1_co... (Conv2D)	(None, 16, 16, 128)	57,344	conv3_block11_
conv3_block11_1_bn (BatchNormalizatio...	(None, 16, 16, 128)	512	conv3_block11_
conv3_block11_1_re... (Activation)	(None, 16, 16, 128)	0	conv3_block11_
conv3_block11_2_co... (Conv2D)	(None, 16, 16, 32)	36,864	conv3_block11_
conv3_block11_conc... (Concatenate)	(None, 16, 16, 480)	0	conv3_block10_ conv3_block11_
conv3_block12_0_bn (BatchNormalizatio...	(None, 16, 16, 480)	1,920	conv3_block11_
conv3_block12_0_re... (Activation)	(None, 16, 16, 480)	0	conv3_block12_

conv3_block12_1_conv (Conv2D)	(None, 16, 16, 128)	61,440	conv3_block12_1_conv
conv3_block12_1_bn (BatchNormalizatio...	(None, 16, 16, 128)	512	conv3_block12_1_bn
conv3_block12_1_relu (Activation)	(None, 16, 16, 128)	0	conv3_block12_1_relu
conv3_block12_2_conv (Conv2D)	(None, 16, 16, 32)	36,864	conv3_block12_2_conv
conv3_block12_concat (Concatenate)	(None, 16, 16, 512)	0	conv3_block11_conv conv3_block12_2_conv
pool3_bn (BatchNormalizatio...	(None, 16, 16, 512)	2,048	conv3_block12_2_conv
pool3_relu (Activation)	(None, 16, 16, 512)	0	pool3_bn[0][0]
pool3_conv (Conv2D)	(None, 16, 16, 256)	131,072	pool3_relu[0][0]
pool3_pool (AveragePooling2D)	(None, 8, 8, 256)	0	pool3_conv[0][0]
conv4_block1_0_bn (BatchNormalizatio...	(None, 8, 8, 256)	1,024	pool3_pool[0][0]
conv4_block1_0_relu (Activation)	(None, 8, 8, 256)	0	conv4_block1_0_bn
conv4_block1_1_conv (Conv2D)	(None, 8, 8, 128)	32,768	conv4_block1_0_relu
conv4_block1_1_bn (BatchNormalizatio...	(None, 8, 8, 128)	512	conv4_block1_1_conv
conv4_block1_1_relu (Activation)	(None, 8, 8, 128)	0	conv4_block1_1_bn
conv4_block1_2_conv (Conv2D)	(None, 8, 8, 32)	36,864	conv4_block1_1_relu
conv4_block1_concat (Concatenate)	(None, 8, 8, 288)	0	conv4_block1_2_conv
conv4_block2_0_bn (BatchNormalizatio...	(None, 8, 8, 288)	1,152	pool3_pool[0][0] conv4_block1_2_conv
conv4_block2_0_relu (Activation)	(None, 8, 8, 288)	0	conv4_block1_0_bn
conv4_block2_1_conv (Conv2D)	(None, 8, 8, 128)	36,864	conv4_block1_c

conv4_block2_1_bn (BatchNormalizatio...	(None, 8, 8, 128)	512	conv4_block2_1
conv4_block2_1_relu (Activation)	(None, 8, 8, 128)	0	conv4_block2_1
conv4_block2_2_conv (Conv2D)	(None, 8, 8, 32)	36,864	conv4_block2_1
conv4_block2_concat (Concatenate)	(None, 8, 8, 320)	0	conv4_block1_c conv4_block2_2
conv4_block3_0_bn (BatchNormalizatio...	(None, 8, 8, 320)	1,280	conv4_block2_c
conv4_block3_0_relu (Activation)	(None, 8, 8, 320)	0	conv4_block3_0
conv4_block3_1_conv (Conv2D)	(None, 8, 8, 128)	40,960	conv4_block3_0
conv4_block3_1_bn (BatchNormalizatio...	(None, 8, 8, 128)	512	conv4_block3_1
conv4_block3_1_relu (Activation)	(None, 8, 8, 128)	0	conv4_block3_1
conv4_block3_2_conv (Conv2D)	(None, 8, 8, 32)	36,864	conv4_block3_1
conv4_block3_concat (Concatenate)	(None, 8, 8, 352)	0	conv4_block2_c conv4_block3_2
conv4_block4_0_bn (BatchNormalizatio...	(None, 8, 8, 352)	1,408	conv4_block3_c
conv4_block4_0_relu (Activation)	(None, 8, 8, 352)	0	conv4_block4_0
conv4_block4_1_conv (Conv2D)	(None, 8, 8, 128)	45,056	conv4_block4_0
conv4_block4_1_bn (BatchNormalizatio...	(None, 8, 8, 128)	512	conv4_block4_1
conv4_block4_1_relu (Activation)	(None, 8, 8, 128)	0	conv4_block4_1
conv4_block4_2_conv (Conv2D)	(None, 8, 8, 32)	36,864	conv4_block4_1
conv4_block4_concat (Concatenate)	(None, 8, 8, 384)	0	conv4_block3_c conv4_block4_2
conv4_block5_0_bn (BatchNormalizatio...	(None, 8, 8, 384)	1,536	conv4_block4_c

conv4_block5_0_relu (Activation)	(None, 8, 8, 384)	0	conv4_block5_0
conv4_block5_1_conv (Conv2D)	(None, 8, 8, 128)	49,152	conv4_block5_0
conv4_block5_1_bn (BatchNormalizatio...	(None, 8, 8, 128)	512	conv4_block5_1
conv4_block5_1_relu (Activation)	(None, 8, 8, 128)	0	conv4_block5_1
conv4_block5_2_conv (Conv2D)	(None, 8, 8, 32)	36,864	conv4_block5_1
conv4_block5_concat (Concatenate)	(None, 8, 8, 416)	0	conv4_block4_c conv4_block5_2
conv4_block6_0_bn (BatchNormalizatio...	(None, 8, 8, 416)	1,664	conv4_block5_c
conv4_block6_0_relu (Activation)	(None, 8, 8, 416)	0	conv4_block6_0
conv4_block6_1_conv (Conv2D)	(None, 8, 8, 128)	53,248	conv4_block6_0
conv4_block6_1_bn (BatchNormalizatio...	(None, 8, 8, 128)	512	conv4_block6_1
conv4_block6_1_relu (Activation)	(None, 8, 8, 128)	0	conv4_block6_1
conv4_block6_2_conv (Conv2D)	(None, 8, 8, 32)	36,864	conv4_block6_1
conv4_block6_concat (Concatenate)	(None, 8, 8, 448)	0	conv4_block5_c conv4_block6_2
conv4_block7_0_bn (BatchNormalizatio...	(None, 8, 8, 448)	1,792	conv4_block6_c
conv4_block7_0_relu (Activation)	(None, 8, 8, 448)	0	conv4_block7_0
conv4_block7_1_conv (Conv2D)	(None, 8, 8, 128)	57,344	conv4_block7_0
conv4_block7_1_bn (BatchNormalizatio...	(None, 8, 8, 128)	512	conv4_block7_1
conv4_block7_1_relu (Activation)	(None, 8, 8, 128)	0	conv4_block7_1
conv4_block7_2_conv (Conv2D)	(None, 8, 8, 32)	36,864	conv4_block7_1

conv4_block7_concat (Concatenate)	(None, 8, 8, 480)	0	conv4_block6_c conv4_block7_2
conv4_block8_0_bn (BatchNormalizatio...	(None, 8, 8, 480)	1,920	conv4_block7_c
conv4_block8_0_relu (Activation)	(None, 8, 8, 480)	0	conv4_block8_0
conv4_block8_1_conv (Conv2D)	(None, 8, 8, 128)	61,440	conv4_block8_0
conv4_block8_1_bn (BatchNormalizatio...	(None, 8, 8, 128)	512	conv4_block8_1
conv4_block8_1_relu (Activation)	(None, 8, 8, 128)	0	conv4_block8_1
conv4_block8_2_conv (Conv2D)	(None, 8, 8, 32)	36,864	conv4_block8_1
conv4_block8_concat (Concatenate)	(None, 8, 8, 512)	0	conv4_block7_c conv4_block8_2
conv4_block9_0_bn (BatchNormalizatio...	(None, 8, 8, 512)	2,048	conv4_block8_c
conv4_block9_0_relu (Activation)	(None, 8, 8, 512)	0	conv4_block9_0
conv4_block9_1_conv (Conv2D)	(None, 8, 8, 128)	65,536	conv4_block9_0
conv4_block9_1_bn (BatchNormalizatio...	(None, 8, 8, 128)	512	conv4_block9_1
conv4_block9_1_relu (Activation)	(None, 8, 8, 128)	0	conv4_block9_1
conv4_block9_2_conv (Conv2D)	(None, 8, 8, 32)	36,864	conv4_block9_1
conv4_block9_concat (Concatenate)	(None, 8, 8, 544)	0	conv4_block8_c conv4_block9_2
conv4_block10_0_bn (BatchNormalizatio...	(None, 8, 8, 544)	2,176	conv4_block9_c
conv4_block10_0_re... (Activation)	(None, 8, 8, 544)	0	conv4_block10_
conv4_block10_1_co... (Conv2D)	(None, 8, 8, 128)	69,632	conv4_block10_
conv4_block10_1_bn (BatchNormalizatio...	(None, 8, 8, 128)	512	conv4_block10_

conv4_block10_1_re... (Activation)	(None, 8, 8, 128)	0	conv4_block10_
conv4_block10_2_co... (Conv2D)	(None, 8, 8, 32)	36,864	conv4_block10_
conv4_block10_conc... (Concatenate)	(None, 8, 8, 576)	0	conv4_block9_c conv4_block10_
conv4_block11_0_bn (BatchNormalizatio...	(None, 8, 8, 576)	2,304	conv4_block10_
conv4_block11_0_re... (Activation)	(None, 8, 8, 576)	0	conv4_block11_
conv4_block11_1_co... (Conv2D)	(None, 8, 8, 128)	73,728	conv4_block11_
conv4_block11_1_bn (BatchNormalizatio...	(None, 8, 8, 128)	512	conv4_block11_
conv4_block11_1_re... (Activation)	(None, 8, 8, 128)	0	conv4_block11_
conv4_block11_2_co... (Conv2D)	(None, 8, 8, 32)	36,864	conv4_block11_
conv4_block11_conc... (Concatenate)	(None, 8, 8, 608)	0	conv4_block10_ conv4_block11_
conv4_block12_0_bn (BatchNormalizatio...	(None, 8, 8, 608)	2,432	conv4_block11_
conv4_block12_0_re... (Activation)	(None, 8, 8, 608)	0	conv4_block12_
conv4_block12_1_co... (Conv2D)	(None, 8, 8, 128)	77,824	conv4_block12_
conv4_block12_1_bn (BatchNormalizatio...	(None, 8, 8, 128)	512	conv4_block12_
conv4_block12_1_re... (Activation)	(None, 8, 8, 128)	0	conv4_block12_
conv4_block12_2_co... (Conv2D)	(None, 8, 8, 32)	36,864	conv4_block12_
conv4_block12_conc... (Concatenate)	(None, 8, 8, 640)	0	conv4_block11_ conv4_block12_
conv4_block13_0_bn (BatchNormalizatio...	(None, 8, 8, 640)	2,560	conv4_block12_
conv4_block13_0_re... (Activation)	(None, 8, 8, 640)	0	conv4_block13_

conv4_block13_1_conv (Conv2D)	(None, 8, 8, 128)	81,920	conv4_block13_1
conv4_block13_1_bn (BatchNormalizatio...	(None, 8, 8, 128)	512	conv4_block13_1
conv4_block13_1_relu (Activation)	(None, 8, 8, 128)	0	conv4_block13_1
conv4_block13_2_conv (Conv2D)	(None, 8, 8, 32)	36,864	conv4_block13_2
conv4_block13_concat (Concatenate)	(None, 8, 8, 672)	0	conv4_block12_conv4_block13_2
conv4_block14_0_bn (BatchNormalizatio...	(None, 8, 8, 672)	2,688	conv4_block13_2
conv4_block14_0_relu (Activation)	(None, 8, 8, 672)	0	conv4_block14_0
conv4_block14_1_conv (Conv2D)	(None, 8, 8, 128)	86,016	conv4_block14_0
conv4_block14_1_bn (BatchNormalizatio...	(None, 8, 8, 128)	512	conv4_block14_1
conv4_block14_1_relu (Activation)	(None, 8, 8, 128)	0	conv4_block14_1
conv4_block14_2_conv (Conv2D)	(None, 8, 8, 32)	36,864	conv4_block14_1
conv4_block14_concat (Concatenate)	(None, 8, 8, 704)	0	conv4_block13_conv4_block14_2
conv4_block15_0_bn (BatchNormalizatio...	(None, 8, 8, 704)	2,816	conv4_block14_2
conv4_block15_0_relu (Activation)	(None, 8, 8, 704)	0	conv4_block15_0
conv4_block15_1_conv (Conv2D)	(None, 8, 8, 128)	90,112	conv4_block15_0
conv4_block15_1_bn (BatchNormalizatio...	(None, 8, 8, 128)	512	conv4_block15_1
conv4_block15_1_relu (Activation)	(None, 8, 8, 128)	0	conv4_block15_1
conv4_block15_2_conv (Conv2D)	(None, 8, 8, 32)	36,864	conv4_block15_1
conv4_block15_concat (Concatenate)	(None, 8, 8, 736)	0	conv4_block14_conv4_block15_2

conv4_block16_0_bn (BatchNormalizatio...	(None, 8, 8, 736)	2,944	conv4_block15_
conv4_block16_0_re... (Activation)	(None, 8, 8, 736)	0	conv4_block16_
conv4_block16_1_co... (Conv2D)	(None, 8, 8, 128)	94,208	conv4_block16_
conv4_block16_1_bn (BatchNormalizatio...	(None, 8, 8, 128)	512	conv4_block16_
conv4_block16_1_re... (Activation)	(None, 8, 8, 128)	0	conv4_block16_
conv4_block16_2_co... (Conv2D)	(None, 8, 8, 32)	36,864	conv4_block16_
conv4_block16_conc... (Concatenate)	(None, 8, 8, 768)	0	conv4_block15_ conv4_block16_
conv4_block17_0_bn (BatchNormalizatio...	(None, 8, 8, 768)	3,072	conv4_block16_
conv4_block17_0_re... (Activation)	(None, 8, 8, 768)	0	conv4_block17_
conv4_block17_1_co... (Conv2D)	(None, 8, 8, 128)	98,304	conv4_block17_
conv4_block17_1_bn (BatchNormalizatio...	(None, 8, 8, 128)	512	conv4_block17_
conv4_block17_1_re... (Activation)	(None, 8, 8, 128)	0	conv4_block17_
conv4_block17_2_co... (Conv2D)	(None, 8, 8, 32)	36,864	conv4_block17_
conv4_block17_conc... (Concatenate)	(None, 8, 8, 800)	0	conv4_block16_ conv4_block17_
conv4_block18_0_bn (BatchNormalizatio...	(None, 8, 8, 800)	3,200	conv4_block17_
conv4_block18_0_re... (Activation)	(None, 8, 8, 800)	0	conv4_block18_
conv4_block18_1_co... (Conv2D)	(None, 8, 8, 128)	102,400	conv4_block18_
conv4_block18_1_bn (BatchNormalizatio...	(None, 8, 8, 128)	512	conv4_block18_
conv4_block18_1_re... (Activation)	(None, 8, 8, 128)	0	conv4_block18_

conv4_block18_2_co... (Conv2D)	(None, 8, 8, 32)	36,864	conv4_block18_
conv4_block18_conc... (Concatenate)	(None, 8, 8, 832)	0	conv4_block17_ conv4_block18_
conv4_block19_0_bn (BatchNormalizatio...	(None, 8, 8, 832)	3,328	conv4_block18_
conv4_block19_0_re... (Activation)	(None, 8, 8, 832)	0	conv4_block19_
conv4_block19_1_co... (Conv2D)	(None, 8, 8, 128)	106,496	conv4_block19_
conv4_block19_1_bn (BatchNormalizatio...	(None, 8, 8, 128)	512	conv4_block19_
conv4_block19_1_re... (Activation)	(None, 8, 8, 128)	0	conv4_block19_
conv4_block19_2_co... (Conv2D)	(None, 8, 8, 32)	36,864	conv4_block19_
conv4_block19_conc... (Concatenate)	(None, 8, 8, 864)	0	conv4_block18_ conv4_block19_
conv4_block20_0_bn (BatchNormalizatio...	(None, 8, 8, 864)	3,456	conv4_block19_
conv4_block20_0_re... (Activation)	(None, 8, 8, 864)	0	conv4_block20_
conv4_block20_1_co... (Conv2D)	(None, 8, 8, 128)	110,592	conv4_block20_
conv4_block20_1_bn (BatchNormalizatio...	(None, 8, 8, 128)	512	conv4_block20_
conv4_block20_1_re... (Activation)	(None, 8, 8, 128)	0	conv4_block20_
conv4_block20_2_co... (Conv2D)	(None, 8, 8, 32)	36,864	conv4_block20_
conv4_block20_conc... (Concatenate)	(None, 8, 8, 896)	0	conv4_block19_ conv4_block20_
conv4_block21_0_bn (BatchNormalizatio...	(None, 8, 8, 896)	3,584	conv4_block20_
conv4_block21_0_re... (Activation)	(None, 8, 8, 896)	0	conv4_block21_
conv4_block21_1_co... (Conv2D)	(None, 8, 8, 128)	114,688	conv4_block21_

conv4_block21_1_bn (BatchNormalizatio...	(None, 8, 8, 128)	512	conv4_block21_
conv4_block21_1_re... (Activation)	(None, 8, 8, 128)	0	conv4_block21_
conv4_block21_2_co... (Conv2D)	(None, 8, 8, 32)	36,864	conv4_block21_
conv4_block21_conc... (Concatenate)	(None, 8, 8, 928)	0	conv4_block20_ conv4_block21_
conv4_block22_0_bn (BatchNormalizatio...	(None, 8, 8, 928)	3,712	conv4_block21_
conv4_block22_0_re... (Activation)	(None, 8, 8, 928)	0	conv4_block22_
conv4_block22_1_co... (Conv2D)	(None, 8, 8, 128)	118,784	conv4_block22_
conv4_block22_1_bn (BatchNormalizatio...	(None, 8, 8, 128)	512	conv4_block22_
conv4_block22_1_re... (Activation)	(None, 8, 8, 128)	0	conv4_block22_
conv4_block22_2_co... (Conv2D)	(None, 8, 8, 32)	36,864	conv4_block22_
conv4_block22_conc... (Concatenate)	(None, 8, 8, 960)	0	conv4_block21_ conv4_block22_
conv4_block23_0_bn (BatchNormalizatio...	(None, 8, 8, 960)	3,840	conv4_block22_
conv4_block23_0_re... (Activation)	(None, 8, 8, 960)	0	conv4_block23_
conv4_block23_1_co... (Conv2D)	(None, 8, 8, 128)	122,880	conv4_block23_
conv4_block23_1_bn (BatchNormalizatio...	(None, 8, 8, 128)	512	conv4_block23_
conv4_block23_1_re... (Activation)	(None, 8, 8, 128)	0	conv4_block23_
conv4_block23_2_co... (Conv2D)	(None, 8, 8, 32)	36,864	conv4_block23_
conv4_block23_conc... (Concatenate)	(None, 8, 8, 992)	0	conv4_block22_ conv4_block23_
conv4_block24_0_bn (BatchNormalizatio...	(None, 8, 8, 992)	3,968	conv4_block23_

conv4_block24_0_re... (Activation)	(None, 8, 8, 992)	0	conv4_block24_
conv4_block24_1_co... (Conv2D)	(None, 8, 8, 128)	126,976	conv4_block24_
conv4_block24_1_bn (BatchNormalizatio...	(None, 8, 8, 128)	512	conv4_block24_
conv4_block24_1_re... (Activation)	(None, 8, 8, 128)	0	conv4_block24_
conv4_block24_2_co... (Conv2D)	(None, 8, 8, 32)	36,864	conv4_block24_
conv4_block24_conc... (Concatenate)	(None, 8, 8, 1024)	0	conv4_block23_ conv4_block24_
pool4_bn (BatchNormalizatio...	(None, 8, 8, 1024)	4,096	conv4_block24_
pool4_relu (Activation)	(None, 8, 8, 1024)	0	pool4_bn[0][0]
pool4_conv (Conv2D)	(None, 8, 8, 512)	524,288	pool4_relu[0][
pool4_pool (AveragePooling2D)	(None, 4, 4, 512)	0	pool4_conv[0][
conv5_block1_0_bn (BatchNormalizatio...	(None, 4, 4, 512)	2,048	pool4_pool[0][
conv5_block1_0_relu (Activation)	(None, 4, 4, 512)	0	conv5_block1_0
conv5_block1_1_conv (Conv2D)	(None, 4, 4, 128)	65,536	conv5_block1_0
conv5_block1_1_bn (BatchNormalizatio...	(None, 4, 4, 128)	512	conv5_block1_1
conv5_block1_1_relu (Activation)	(None, 4, 4, 128)	0	conv5_block1_1
conv5_block1_2_conv (Conv2D)	(None, 4, 4, 32)	36,864	conv5_block1_1
conv5_block1_concat (Concatenate)	(None, 4, 4, 544)	0	pool4_pool[0][conv5_block1_2
conv5_block2_0_bn (BatchNormalizatio...	(None, 4, 4, 544)	2,176	conv5_block1_c
conv5_block2_0_relu (Activation)	(None, 4, 4, 544)	0	conv5_block2_0
conv5_block2_1_conv	(None, 4, 4, 128)	69,632	conv5_block2_0

(Conv2D)			
conv5_block2_1_bn (BatchNormalizatio...	(None, 4, 4, 128)	512	conv5_block2_1
conv5_block2_1_relu (Activation)	(None, 4, 4, 128)	0	conv5_block2_1
conv5_block2_2_conv (Conv2D)	(None, 4, 4, 32)	36,864	conv5_block2_1
conv5_block2_concat (Concatenate)	(None, 4, 4, 576)	0	conv5_block1_c conv5_block2_2
conv5_block3_0_bn (BatchNormalizatio...	(None, 4, 4, 576)	2,304	conv5_block2_c
conv5_block3_0_relu (Activation)	(None, 4, 4, 576)	0	conv5_block3_0
conv5_block3_1_conv (Conv2D)	(None, 4, 4, 128)	73,728	conv5_block3_0
conv5_block3_1_bn (BatchNormalizatio...	(None, 4, 4, 128)	512	conv5_block3_1
conv5_block3_1_relu (Activation)	(None, 4, 4, 128)	0	conv5_block3_1
conv5_block3_2_conv (Conv2D)	(None, 4, 4, 32)	36,864	conv5_block3_1
conv5_block3_concat (Concatenate)	(None, 4, 4, 608)	0	conv5_block2_c conv5_block3_2
conv5_block4_0_bn (BatchNormalizatio...	(None, 4, 4, 608)	2,432	conv5_block3_c
conv5_block4_0_relu (Activation)	(None, 4, 4, 608)	0	conv5_block4_0
conv5_block4_1_conv (Conv2D)	(None, 4, 4, 128)	77,824	conv5_block4_0
conv5_block4_1_bn (BatchNormalizatio...	(None, 4, 4, 128)	512	conv5_block4_1
conv5_block4_1_relu (Activation)	(None, 4, 4, 128)	0	conv5_block4_1
conv5_block4_2_conv (Conv2D)	(None, 4, 4, 32)	36,864	conv5_block4_1
conv5_block4_concat (Concatenate)	(None, 4, 4, 640)	0	conv5_block3_c conv5_block4_2
conv5_block5_0_bn	(None, 4, 4, 640)	2,560	conv5_block4_c

(BatchNormalizatio...			
conv5_block5_0_relu (Activation)	(None, 4, 4, 640)	0	conv5_block5_0
conv5_block5_1_conv (Conv2D)	(None, 4, 4, 128)	81,920	conv5_block5_0
conv5_block5_1_bn (BatchNormalizatio...	(None, 4, 4, 128)	512	conv5_block5_1
conv5_block5_1_relu (Activation)	(None, 4, 4, 128)	0	conv5_block5_1
conv5_block5_2_conv (Conv2D)	(None, 4, 4, 32)	36,864	conv5_block5_1
conv5_block5_concat (Concatenate)	(None, 4, 4, 672)	0	conv5_block4_c conv5_block5_2
conv5_block6_0_bn (BatchNormalizatio...	(None, 4, 4, 672)	2,688	conv5_block5_c
conv5_block6_0_relu (Activation)	(None, 4, 4, 672)	0	conv5_block6_0
conv5_block6_1_conv (Conv2D)	(None, 4, 4, 128)	86,016	conv5_block6_0
conv5_block6_1_bn (BatchNormalizatio...	(None, 4, 4, 128)	512	conv5_block6_1
conv5_block6_1_relu (Activation)	(None, 4, 4, 128)	0	conv5_block6_1
conv5_block6_2_conv (Conv2D)	(None, 4, 4, 32)	36,864	conv5_block6_1
conv5_block6_concat (Concatenate)	(None, 4, 4, 704)	0	conv5_block5_c conv5_block6_2
conv5_block7_0_bn (BatchNormalizatio...	(None, 4, 4, 704)	2,816	conv5_block6_c
conv5_block7_0_relu (Activation)	(None, 4, 4, 704)	0	conv5_block7_0
conv5_block7_1_conv (Conv2D)	(None, 4, 4, 128)	90,112	conv5_block7_0
conv5_block7_1_bn (BatchNormalizatio...	(None, 4, 4, 128)	512	conv5_block7_1
conv5_block7_1_relu (Activation)	(None, 4, 4, 128)	0	conv5_block7_1
conv5_block7_2_conv	(None, 4, 4, 32)	36,864	conv5_block7_1

(Conv2D)			
conv5_block7_concat (Concatenate)	(None, 4, 4, 736)	0	conv5_block6_c conv5_block7_2
conv5_block8_0_bn (BatchNormalizatio...	(None, 4, 4, 736)	2,944	conv5_block7_c
conv5_block8_0_relu (Activation)	(None, 4, 4, 736)	0	conv5_block8_0
conv5_block8_1_conv (Conv2D)	(None, 4, 4, 128)	94,208	conv5_block8_0
conv5_block8_1_bn (BatchNormalizatio...	(None, 4, 4, 128)	512	conv5_block8_1
conv5_block8_1_relu (Activation)	(None, 4, 4, 128)	0	conv5_block8_1
conv5_block8_2_conv (Conv2D)	(None, 4, 4, 32)	36,864	conv5_block8_1
conv5_block8_concat (Concatenate)	(None, 4, 4, 768)	0	conv5_block7_c conv5_block8_2
conv5_block9_0_bn (BatchNormalizatio...	(None, 4, 4, 768)	3,072	conv5_block8_c
conv5_block9_0_relu (Activation)	(None, 4, 4, 768)	0	conv5_block9_0
conv5_block9_1_conv (Conv2D)	(None, 4, 4, 128)	98,304	conv5_block9_0
conv5_block9_1_bn (BatchNormalizatio...	(None, 4, 4, 128)	512	conv5_block9_1
conv5_block9_1_relu (Activation)	(None, 4, 4, 128)	0	conv5_block9_1
conv5_block9_2_conv (Conv2D)	(None, 4, 4, 32)	36,864	conv5_block9_1
conv5_block9_concat (Concatenate)	(None, 4, 4, 800)	0	conv5_block8_c conv5_block9_2
conv5_block10_0_bn (BatchNormalizatio...	(None, 4, 4, 800)	3,200	conv5_block9_c
conv5_block10_0_re... (Activation)	(None, 4, 4, 800)	0	conv5_block10_
conv5_block10_1_co... (Conv2D)	(None, 4, 4, 128)	102,400	conv5_block10_
conv5_block10_1_bn	(None, 4, 4, 128)	512	conv5_block10_

(BatchNormalizatio...			
conv5_block10_1_re... (Activation)	(None, 4, 4, 128)	0	conv5_block10_
conv5_block10_2_co... (Conv2D)	(None, 4, 4, 32)	36,864	conv5_block10_
conv5_block10_conc... (Concatenate)	(None, 4, 4, 832)	0	conv5_block9_c conv5_block10_
conv5_block11_0_bn (BatchNormalizatio...	(None, 4, 4, 832)	3,328	conv5_block10_
conv5_block11_0_re... (Activation)	(None, 4, 4, 832)	0	conv5_block11_
conv5_block11_1_co... (Conv2D)	(None, 4, 4, 128)	106,496	conv5_block11_
conv5_block11_1_bn (BatchNormalizatio...	(None, 4, 4, 128)	512	conv5_block11_
conv5_block11_1_re... (Activation)	(None, 4, 4, 128)	0	conv5_block11_
conv5_block11_2_co... (Conv2D)	(None, 4, 4, 32)	36,864	conv5_block11_
conv5_block11_conc... (Concatenate)	(None, 4, 4, 864)	0	conv5_block10_ conv5_block11_
conv5_block12_0_bn (BatchNormalizatio...	(None, 4, 4, 864)	3,456	conv5_block11_
conv5_block12_0_re... (Activation)	(None, 4, 4, 864)	0	conv5_block12_
conv5_block12_1_co... (Conv2D)	(None, 4, 4, 128)	110,592	conv5_block12_
conv5_block12_1_bn (BatchNormalizatio...	(None, 4, 4, 128)	512	conv5_block12_
conv5_block12_1_re... (Activation)	(None, 4, 4, 128)	0	conv5_block12_
conv5_block12_2_co... (Conv2D)	(None, 4, 4, 32)	36,864	conv5_block12_
conv5_block12_conc... (Concatenate)	(None, 4, 4, 896)	0	conv5_block11_ conv5_block12_
conv5_block13_0_bn (BatchNormalizatio...	(None, 4, 4, 896)	3,584	conv5_block12_
conv5_block13_0_re...	(None, 4, 4, 896)	0	conv5_block13_

(Activation)			
conv5_block13_1_conv (Conv2D)	(None, 4, 4, 128)	114,688	conv5_block13_
conv5_block13_1_bn (BatchNormalizatio...	(None, 4, 4, 128)	512	conv5_block13_
conv5_block13_1_relu (Activation)	(None, 4, 4, 128)	0	conv5_block13_
conv5_block13_2_conv (Conv2D)	(None, 4, 4, 32)	36,864	conv5_block13_
conv5_block13_concat (Concatenate)	(None, 4, 4, 928)	0	conv5_block12_ conv5_block13_
conv5_block14_0_bn (BatchNormalizatio...	(None, 4, 4, 928)	3,712	conv5_block13_
conv5_block14_0_relu (Activation)	(None, 4, 4, 928)	0	conv5_block14_
conv5_block14_1_conv (Conv2D)	(None, 4, 4, 128)	118,784	conv5_block14_
conv5_block14_1_bn (BatchNormalizatio...	(None, 4, 4, 128)	512	conv5_block14_
conv5_block14_1_relu (Activation)	(None, 4, 4, 128)	0	conv5_block14_
conv5_block14_2_conv (Conv2D)	(None, 4, 4, 32)	36,864	conv5_block14_
conv5_block14_concat (Concatenate)	(None, 4, 4, 960)	0	conv5_block13_ conv5_block14_
conv5_block15_0_bn (BatchNormalizatio...	(None, 4, 4, 960)	3,840	conv5_block14_
conv5_block15_0_relu (Activation)	(None, 4, 4, 960)	0	conv5_block15_
conv5_block15_1_conv (Conv2D)	(None, 4, 4, 128)	122,880	conv5_block15_
conv5_block15_1_bn (BatchNormalizatio...	(None, 4, 4, 128)	512	conv5_block15_
conv5_block15_1_relu (Activation)	(None, 4, 4, 128)	0	conv5_block15_
conv5_block15_2_conv (Conv2D)	(None, 4, 4, 32)	36,864	conv5_block15_
conv5_block15_concat	(None, 4, 4, 992)	0	conv5_block14_

(Concatenate)			conv5_block15_
conv5_block16_0_bn (BatchNormalizatio...	(None, 4, 4, 992)	3,968	conv5_block15_
conv5_block16_0_re... (Activation)	(None, 4, 4, 992)	0	conv5_block16_
conv5_block16_1_co... (Conv2D)	(None, 4, 4, 128)	126,976	conv5_block16_
conv5_block16_1_bn (BatchNormalizatio...	(None, 4, 4, 128)	512	conv5_block16_
conv5_block16_1_re... (Activation)	(None, 4, 4, 128)	0	conv5_block16_
conv5_block16_2_co... (Conv2D)	(None, 4, 4, 32)	36,864	conv5_block16_
conv5_block16_conc... (Concatenate)	(None, 4, 4, 1024)	0	conv5_block15_ conv5_block16_
bn (BatchNormalizatio...	(None, 4, 4, 1024)	4,096	conv5_block16_
relu (Activation)	(None, 4, 4, 1024)	0	bn[0][0]
global_average_poo... (GlobalAveragePool...	(None, 1024)	0	relu[0][0]
dense_2 (Dense)	(None, 512)	524,800	global_average
dropout_1 (Dropout)	(None, 512)	0	dense_2[0][0]
dense_3 (Dense)	(None, 2)	1,026	dropout_1[0][0]

Total params: 7,563,330 (28.85 MB)

Trainable params: 7,479,682 (28.53 MB)

Non-trainable params: 83,648 (326.75 KB)

```

In [41]: model = Sequential()

# Input Layer
model.add(Conv2D(32,(3,3),activation='relu',input_shape = (image_size,image_size)))
model.add(BatchNormalization())
model.add(MaxPooling2D(pool_size=(2,2)))
model.add(Dropout(0.2))

# Block 1
model.add(Conv2D(64,(3,3),activation='relu'))
model.add(BatchNormalization())
model.add(MaxPooling2D(pool_size=(2,2)))
model.add(Dropout(0.2))

# Block 2
model.add(Conv2D(128,(3,3),activation='relu'))
model.add(BatchNormalization())
model.add(MaxPooling2D(pool_size=(2,2)))
model.add(Dropout(0.2))

# Block 3
model.add(Conv2D(256,(3,3),activation='relu'))
model.add(BatchNormalization())
model.add(MaxPooling2D(pool_size=(2,2)))
model.add(Dropout(0.2))

# Fully Connected layers
model.add(Flatten())
model.add(Dense(512,activation='relu'))
model.add(BatchNormalization())
model.add(Dropout(0.2))

# Output layer
model.add(Dense(2,activation='softmax'))

model.summary()

```

/home/student/anaconda3/lib/python3.11/site-packages/keras/src/layers/convolutional/base_conv.py:107: UserWarning: Do not pass an `input\_shape`/`input\_dim` argument to a layer. When using Sequential models, prefer using an `Input(shape)` object as the first layer in the model instead.

```

    super().__init__(activity_regularizer=activity_regularizer, **kwargs)
Model: "sequential"

```

Layer (type)	Output Shape	Para
conv2d (Conv2D)	(None, 126, 126, 32)	
batch_normalization (BatchNormalization)	(None, 126, 126, 32)	
max_pooling2d (MaxPooling2D)	(None, 63, 63, 32)	
dropout_2 (Dropout)	(None, 63, 63, 32)	
conv2d_1 (Conv2D)	(None, 61, 61, 64)	18,
batch_normalization_1 (BatchNormalization)	(None, 61, 61, 64)	
max_pooling2d_1 (MaxPooling2D)	(None, 30, 30, 64)	
dropout_3 (Dropout)	(None, 30, 30, 64)	
conv2d_2 (Conv2D)	(None, 28, 28, 128)	73,
batch_normalization_2 (BatchNormalization)	(None, 28, 28, 128)	
max_pooling2d_2 (MaxPooling2D)	(None, 14, 14, 128)	
dropout_4 (Dropout)	(None, 14, 14, 128)	
conv2d_3 (Conv2D)	(None, 12, 12, 256)	295,
batch_normalization_3 (BatchNormalization)	(None, 12, 12, 256)	1,
max_pooling2d_3 (MaxPooling2D)	(None, 6, 6, 256)	
dropout_5 (Dropout)	(None, 6, 6, 256)	
flatten (Flatten)	(None, 9216)	
dense_4 (Dense)	(None, 512)	4,719,
batch_normalization_4 (BatchNormalization)	(None, 512)	2,
dropout_6 (Dropout)	(None, 512)	
dense_5 (Dense)	(None, 2)	1,

Total params: 5,112,514 (19.50 MB)

Trainable params: 5,110,530 (19.50 MB)

Non-trainable params: 1,984 (7.75 KB)

Callbacks

ReduceLROnPlateau: Decreases the learning rate when a monitored metric shows no further improvement during training.

EarlyStopping: Halts the training process if the monitored metric does not improve for a specified number of epochs.

```
In [42]: learning_rate_reduction = ReduceLROnPlateau(monitor = 'val_accuracy',
                                                    patience=2,
                                                    factor=0.5,
                                                    min_lr = 0.00001,
                                                    verbose = 1)

early_stopping = EarlyStopping(monitor='val_loss',patience= 3,restore_best_we
```

Compile the model

Finally, we will compile the model, focusing on three key elements: Optimizer, Loss, and Metrics.

Optimizer: To reduce the cost function, various methods like gradient descent and stochastic gradient descent are used. In this case, we are using the default optimizer, Adam.

Loss: Neural networks aim to minimize the loss to improve performance. We can use different loss functions like 'categorical_crossentropy' or 'binary_crossentropy'. Here, 'binary_crossentropy' is applied.

Metrics: Metrics are used to assess the performance of the model. Commonly used metrics include accuracy, but other measures can be chosen as well.

```
In [43]: model.compile(optimizer='adam',loss='binary_crossentropy',metrics=['accuracy
```

Fit the model

We will now proceed to train the compiled model using the training iterator (train_generator) and validate the model using the validation iterator (val_generator) during training.

It's important to define the number of steps for both the training and validation iterators.

This represents the number of batches per epoch, which is calculated by dividing the total number of images in the training and validation directories by the batch size (32).

The model will be trained for 5 epochs.

```
In [45]: cat_dog = model.fit(train_generator,
                             validation_data = val_generator,
                             callbacks=[early_stopping, learning_rate_reduction],
                             epochs = 5,
                             # steps_per_epoch = len(train_generator),
                             # validation_steps = len(val_generaotor),
                             )
```

Epoch 1/5

625/625 ————— **819s** 1s/step - accuracy: 0.7404 - loss: 0.5265
 - val_accuracy: 0.5952 - val_loss: 0.7542 - learning_rate: 0.0010

Epoch 2/5

625/625 ————— **1004s** 2s/step - accuracy: 0.7853 - loss: 0.4583
 - val_accuracy: 0.7696 - val_loss: 0.4835 - learning_rate: 0.0010

Epoch 3/5

625/625 ————— **1065s** 2s/step - accuracy: 0.8282 - loss: 0.3868
 - val_accuracy: 0.8448 - val_loss: 0.3452 - learning_rate: 0.0010

Epoch 4/5

625/625 ————— **1765s** 3s/step - accuracy: 0.8449 - loss: 0.3464
 - val_accuracy: 0.8532 - val_loss: 0.3222 - learning_rate: 0.0010

Epoch 5/5

625/625 ————— **784s** 1s/step - accuracy: 0.8633 - loss: 0.3199
 - val_accuracy: 0.8444 - val_loss: 0.3559 - learning_rate: 0.0010

Plot the results

```
In [46]: # plots for accuracy and Loss with epochs

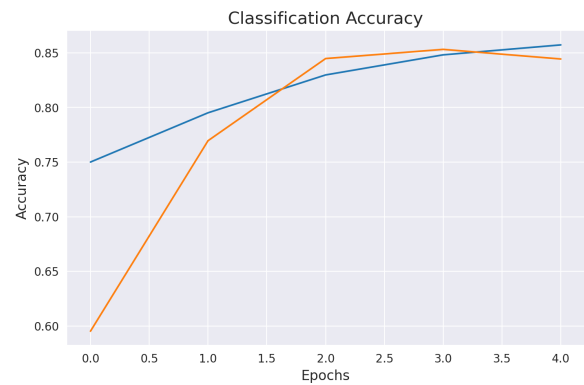
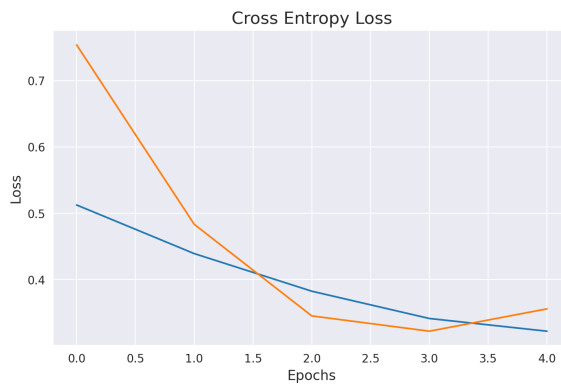
error = pd.DataFrame(cat_dog.history)

plt.figure(figsize=(18,5),dpi=200)
sns.set_style('darkgrid')

plt.subplot(121)
plt.title('Cross Entropy Loss',fontsize=15)
plt.xlabel('Epochs',fontsize=12)
plt.ylabel('Loss',fontsize=12)
plt.plot(error['loss'])
plt.plot(error['val_loss'])

plt.subplot(122)
plt.title('Classification Accuracy',fontsize=15)
plt.xlabel('Epochs',fontsize=12)
plt.ylabel('Accuracy',fontsize=12)
plt.plot(error['accuracy'])
plt.plot(error['val_accuracy'])

plt.show()
```



Evaluation

We will assess the accuracy and loss of both the training and validation datasets to evaluate the model's performance and ensure it is learning effectively from the provided data during training.

```
In [47]: # Evaluvate for train generator
loss,acc = model.evaluate(train_generator,batch_size = bat_size, verbose = 0)

print('The accuracy of the model for training data is:',acc*100)
print('The Loss of the model for training data is:',loss)

# Evaluvate for validation generator
loss,acc = model.evaluate(val_generator,batch_size = bat_size, verbose = 0)

print('The accuracy of the model for validation data is:',acc*100)
print('The Loss of the model for validation data is:',loss)
```

```
The accuracy of the model for training data is: 85.03000140190125
The Loss of the model for training data is: 0.3420724868774414
The accuracy of the model for validation data is: 85.32000184059143
The Loss of the model for validation data is: 0.32220104336738586
```

We will assess the accuracy and loss of both the training and validation datasets to evaluate the model's performance and ensure it is learning effectively from the provided data during training.

```
In [48]: # Save the Model
model.save("model.h5")
```

```
WARNING:absl:You are saving your model as an HDF5 file via `model.save()` or
`keras.saving.save_model(model)`. This file format is considered legacy. We
recommend using instead the native Keras format, e.g. `model.save('my_model.
keras')` or `keras.saving.save_model(model, 'my_model.keras')`.
```

Prediction

In the final step, we will store the trained model for later use, allowing us to reuse it without retraining when needed in the future.

```
In [49]: # prediction
result = model.predict(test_generator, batch_size = bat_size, verbose = 0)

y_pred = np.argmax(result, axis = 1)

y_true = test_generator.labels

# Evaluvate
loss, acc = model.evaluate(test_generator, batch_size = bat_size, verbose = 0)

print('The accuracy of the model for testing data is:', acc*100)
print('The Loss of the model for testing data is:', loss)
```

The accuracy of the model for testing data is: 86.91999912261963
The Loss of the model for testing data is: 0.3089852035045624

Classification report

```
In [50]: labels = ['Cat', 'Dog']
print(classification_report(y_true, y_pred, target_names=labels))
```

	precision	recall	f1-score	support
Cat	0.84	0.92	0.87	1250
Dog	0.91	0.82	0.86	1250
accuracy			0.87	2500
macro avg	0.87	0.87	0.87	2500
weighted avg	0.87	0.87	0.87	2500

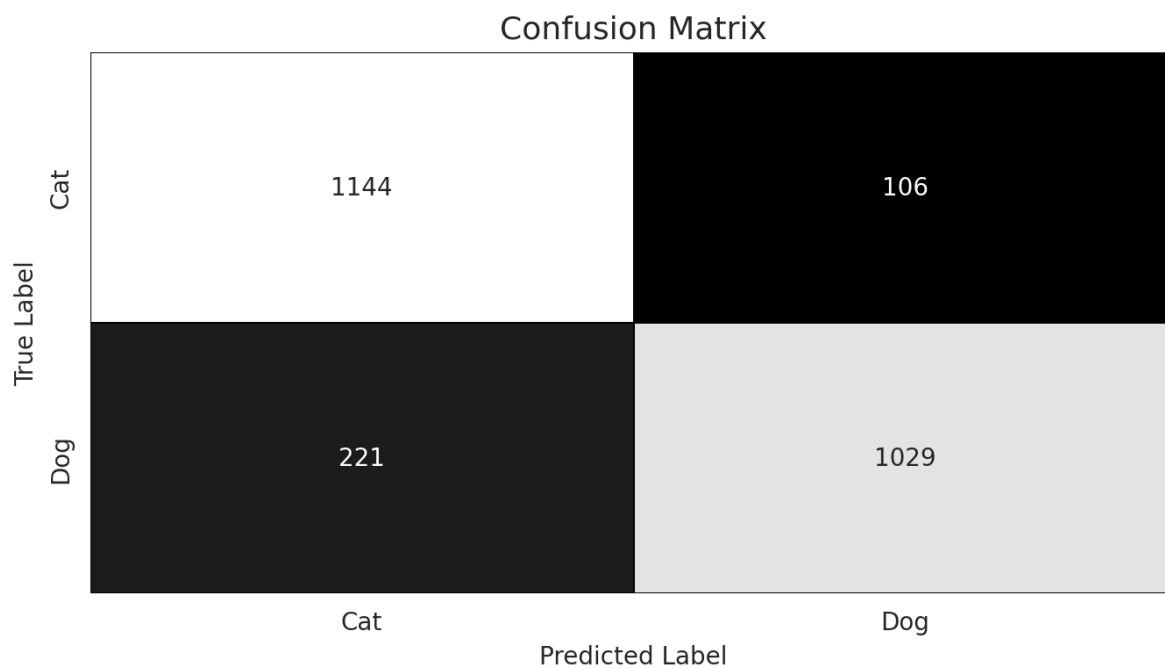
confusioin matrix

```
In [52]: confusion_mtx = confusion_matrix(y_true, y_pred)

f, ax = plt.subplots(figsize = (8,4), dpi=200)
sns.heatmap(confusion_mtx, annot=True, linewidths=0.1, cmap = "gist_yarg_r",

plt.xlabel("Predicted Label", fontsize=10)
plt.ylabel("True Label", fontsize=10)
plt.title("Confusion Matrix", fontsize=13)

plt.show()
```



submission

Make predictions on test data for submission.

In [129...

```

import os
import pandas as pd
import numpy as np
from tensorflow.keras.preprocessing.image import ImageDataGenerator
from tensorflow.keras.models import load_model

# Define the size for the images
size = (128, 128)

# Initialize ImageDataGenerator for test data
test_datagen = ImageDataGenerator(rescale=1./255)

# Set the batch size (ensure that bat_size is defined or set a value here)
bat_size = 32 # Example value for batch size

# Loading filenames into a dataframe
test_dir = "/home/student/Documents/Review block/Review block 4/Week 2/Assign
filenames = os.listdir(test_dir)

# Check if the filenames are valid image files (e.g., jpg, png)
valid_filenames = [f for f in filenames if f.endswith(('.jpg', '.jpeg', '.png'))]
if not valid_filenames:
    print("No valid image files found in the directory.")
else:
    test_data = pd.DataFrame({"filename": valid_filenames})
    test_data['label'] = 'unknown'

# Create a data generator for test data
test1_idg = test_datagen.flow_from_dataframe(test_data,
                                              directory=test_dir, # Specify
                                              x_col="filename",
                                              y_col='label',
                                              batch_size=bat_size,
                                              target_size=size,
                                              shuffle=False)

# Load the pre-trained model (ensure you have a model loaded or use your
model_path = 'model.h5' # Update this path to your actual model path
if not os.path.exists(model_path):
    print(f"Model file not found at {model_path}")
else:
    model = load_model(model_path)

# Test Prediction
test1_predict = model.predict(test1_idg, verbose=0)

# Get the index of the maximum predicted class for each image
test1_predict_argmax = np.argmax(test1_predict, axis=1)

y_test_pred = test1_predict_argmax

# Assign predictions to the dataframe
test_data['label'] = y_test_pred

# Map the predicted labels to the class names
label_mapping = {0: 'cat', 1: 'dog'}
test_data['label'] = test_data['label'].map(label_mapping)

# Show the first few rows of the dataframe with labels

```

```
# Show the first few rows of the dataframe with labels
print(test_data.head())

# CSV file output for submission
sub = pd.read_csv('/home/student/Documents/Review block/Review block

# Assign predictions to the submission dataframe
sub['label'] = y_test_pred

# Save the submission dataframe to a CSV file
sub.to_csv('submission.csv', index=True)
```

Found 12500 validated image filenames belonging to 1 classes.

WARNING:absl:Compiled the loaded model, but the compiled metrics have yet to be built. `model.compile\_metrics` will be empty until you train or evaluate the model.

/home/student/anaconda3/lib/python3.11/site-packages/keras/src/trainers/data_adapters/py_dataset_adapter.py:121: UserWarning: Your `PyDataset` class should call `super().\_\_init\_\_(\*\*kwargs)` in its constructor. `\*\*kwargs` can include `workers`, `use\_multiprocessing`, `max\_queue\_size`. Do not pass these arguments to `fit()`, as they will be ignored.

```
self._warn_if_super_not_called()
filename label
0 3889.jpg cat
1 6797.jpg dog
2 12040.jpg dog
3 8338.jpg cat
4 8218.jpg cat
```

Visualize Classified Images

```
In [130... fig, axes = plt.subplots(1, 10, figsize=(20, 4))
for idx in range(10):
    image_path = os.path.join(test_dir, test_data.iloc[idx]['filename'])
    image = Image.open(image_path)
    axes[idx].imshow(image)
    axes[idx].set_title("Label: " + test_data.iloc[idx]['label'])
    axes[idx].axis('off')
plt.show()
```



After analyzing the images, we observe that our model performs well, achieving an accuracy of 87%.

The predictions align closely with actual values, demonstrating reliable and consistent performance.

Conclusion

We successfully developed a deep learning model using a Convolutional Neural Network (CNN) to classify images of cats and dogs, achieving an impressive 87% accuracy.

The model was tested on an independent dataset, and predictions were submitted to evaluate its performance on new data.

This Cat vs Dog classification project highlights the effectiveness of CNNs in image recognition tasks. By accurately distinguishing between cat and dog images,

it demonstrates the power of deep learning in real-world applications.

This project serves as an inspiration for further research into CNNs and their potential across various fields of computer vision and AI.

In []: